



Vaping related lung injuries

Read more about lung injuries from vaping that have suggestive patterns and characteristics. <http://digit.in/sep20-53>



Breast cancer drug

Phase 1 human trials suggest that a new type of breast cancer drug is both non-toxic and effective. <http://digit.in/sep20-54>

cars. The antennas made using MXene are functionally identical to the ones used in consumer cell phones or wireless routers. The antennas could be also used on flexible electronics, which could range from biomedical monitors to smart fabrics integrated with the MXene to various IOT applications. These antennas add barely any weight or bulk to the device, and can transmit or receive as efficiently as the regular antennas. In tests, the MXene antennas performed 50 times better than their graphene counterparts. Most other nanomaterials, on application,



MXene materials absorbing electromagnetic interference

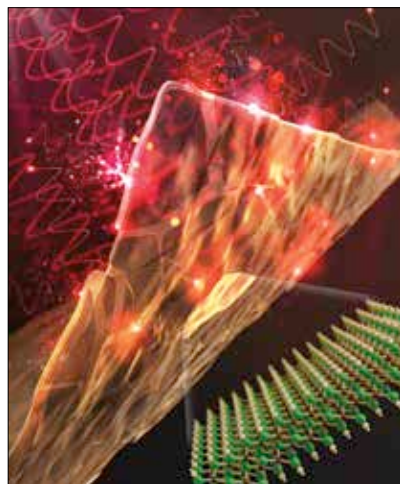
require curing or binders to fix them into place. This is not the case with MXene, as it can simply be sprayed onto to any surface. When combined with 3D printing, it can be a really versatile material.

Another important application that was discovered was the potential to use MXene as a filter for removing urea from blood, which currently requires big and expensive dialysis machines. The MXene worked better at removing the urea from the blood than other known sorbents, in a study conducted in collaboration with many universities around the world, including Symbiosis International University in India. The existing portable dialysis devices still do not work very well when it comes to filtering out urea, and this is the crucial step that is holding back the medical advancement of a wearable kidney, something that could benefit 3 million people around the world.

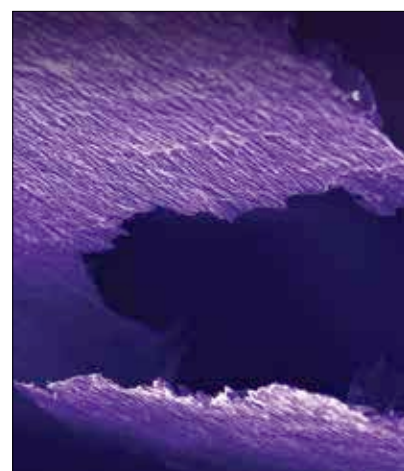
Then in 2019, came another breakthrough. For over a decade, researchers have been trying to figure out how to make conductive ink at scale.

At present, the technology is used to a limited extent, such as making RFID tags or some circuit boards. The promising materials for the same include silver, graphene and gallium. However, they all suffer from the additional curing or binding phase that is required after the printing, to make them stick to the surface. After the application, or printing, a thermal or chemical treatment is required to remove an additive that holds the particles suspended when it is in the ink phase. For MXene, the additive is water or an organic solution that does not require any additional curing, which means that using MXene is somewhat comparable to acrylic paints. The researchers were able to make electronic circuits by adding just concentration of MXene particles in an ink, that was then printed out using a regular inkjet printer. The ink was also tested on materials such as glass and plastic, and performed uniformly well on all of them. The technology promises to allow anyone to rapidly prototype electronic devices at home, apart from the commercial applications.

Consumers are very rarely exposed to electromagnetic interference or EMI, because of all the technology that is used to prevent them in our electronics. Typical approaches involve placing the entire device within a copper shield, or adding a film to individual components. Both



MXene can be used as EM shields for flexible electronics



MXene clay

of these approaches add weight and bulk to the device. Researches have shown that a MXene film was 3-5 times better at blocking EMI than a copper film of the same width. The researchers also found out something surprising about MXene, instead of just reflecting off the EMI, which is what copper does, it actually absorbed the EMI while still protecting the components. This meant that the waste radiation would no longer propagate in the environment.

Recently, there have been advances made in the manufacturing of MXene at scale, and all the benefits of the material could actually be available soon to consumers because of how cheap it is to produce, and the relatively low costs involved. The major benefit that MXene has over other promising 2D nanomaterials, is that it is a subtractive process, not an additive one, where the material has to be assembled atom by atom, a process that can be expensive and time consuming. MXene is one of the few 2D materials that has been proved to be viable for large scale manufacturing. The solubility in water makes it versatile enough to be easily integrated into existing manufacturing processes. A number of electronics companies are currently developing applications for MXene, and you might just hold a device that uses it sooner than you see a widespread consumer application for graphene. **4**